

# CrabTransformer Training Report

Scenario: REAL | Loss: MSE | Lag: 5 Years | Current Spawner: Yes

## — Training Configuration —

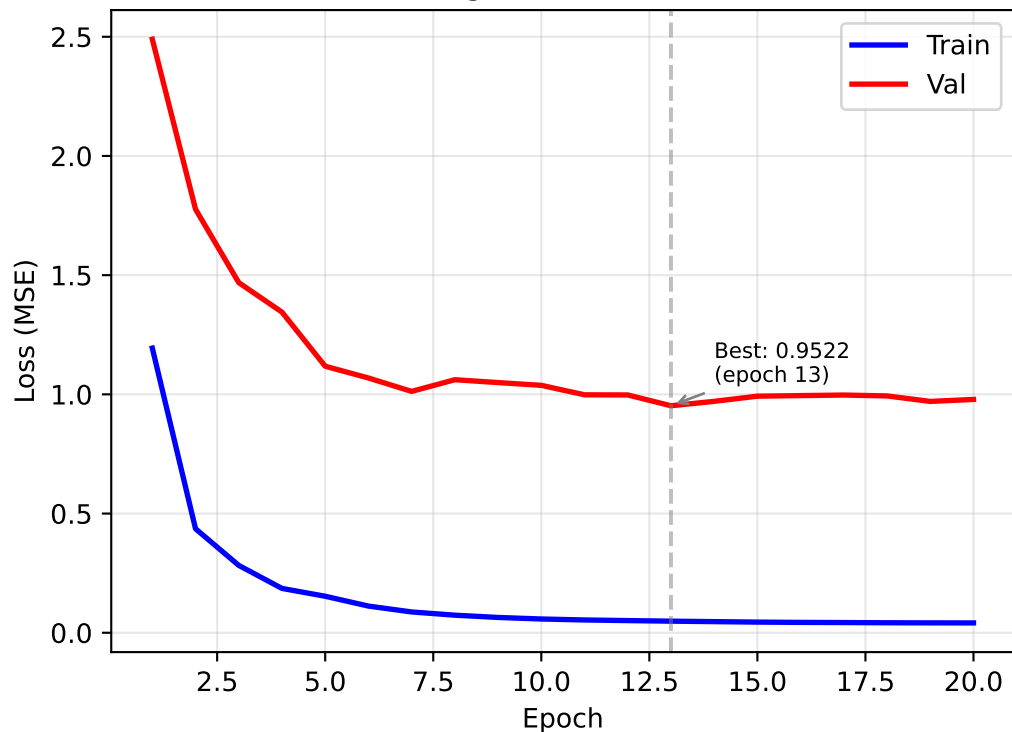
epochs\_run: 20      best\_val\_epoch: 13  
best\_val\_loss: 0.9522      final\_train\_loss: 0.0413  
bias\_correction (MSE): 1.0628  
include\_current\_spawner: True

## — Model Architecture —

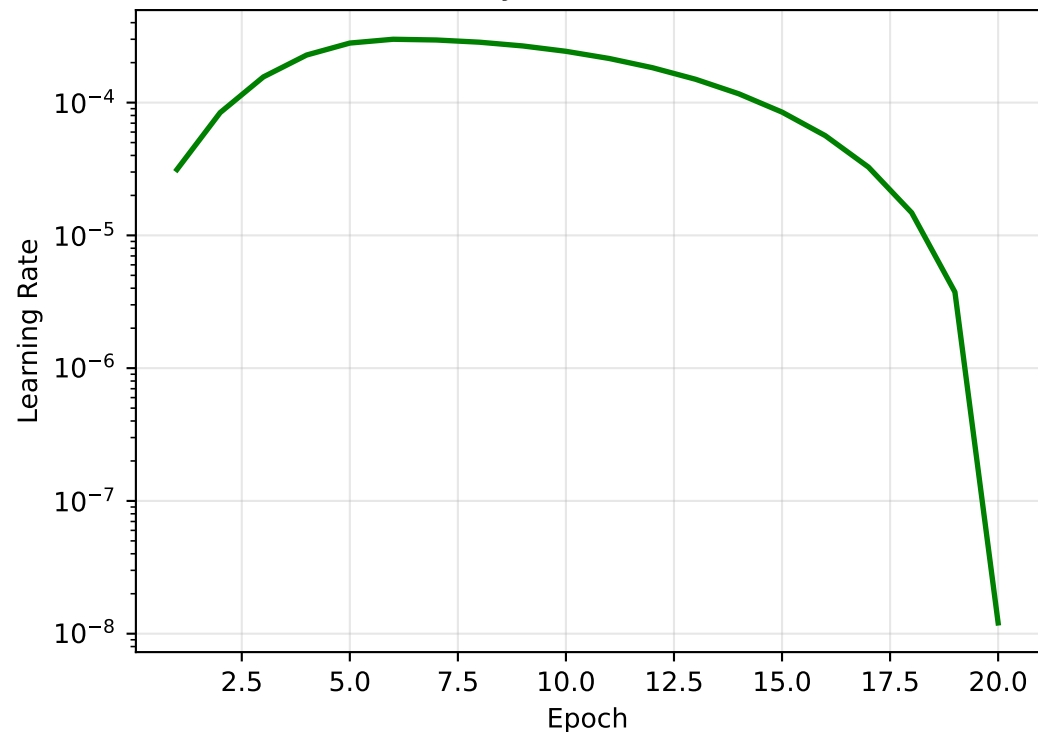
embed\_dim: 128      num\_heads: 8      num\_layers: 6  
d\_ff: 512      dropout: 0.1      patch\_size: 5  
train/val/test years: 21/6/4

# Training Curves: REAL | Lag 5 | MSE | w/ Spawner(t)

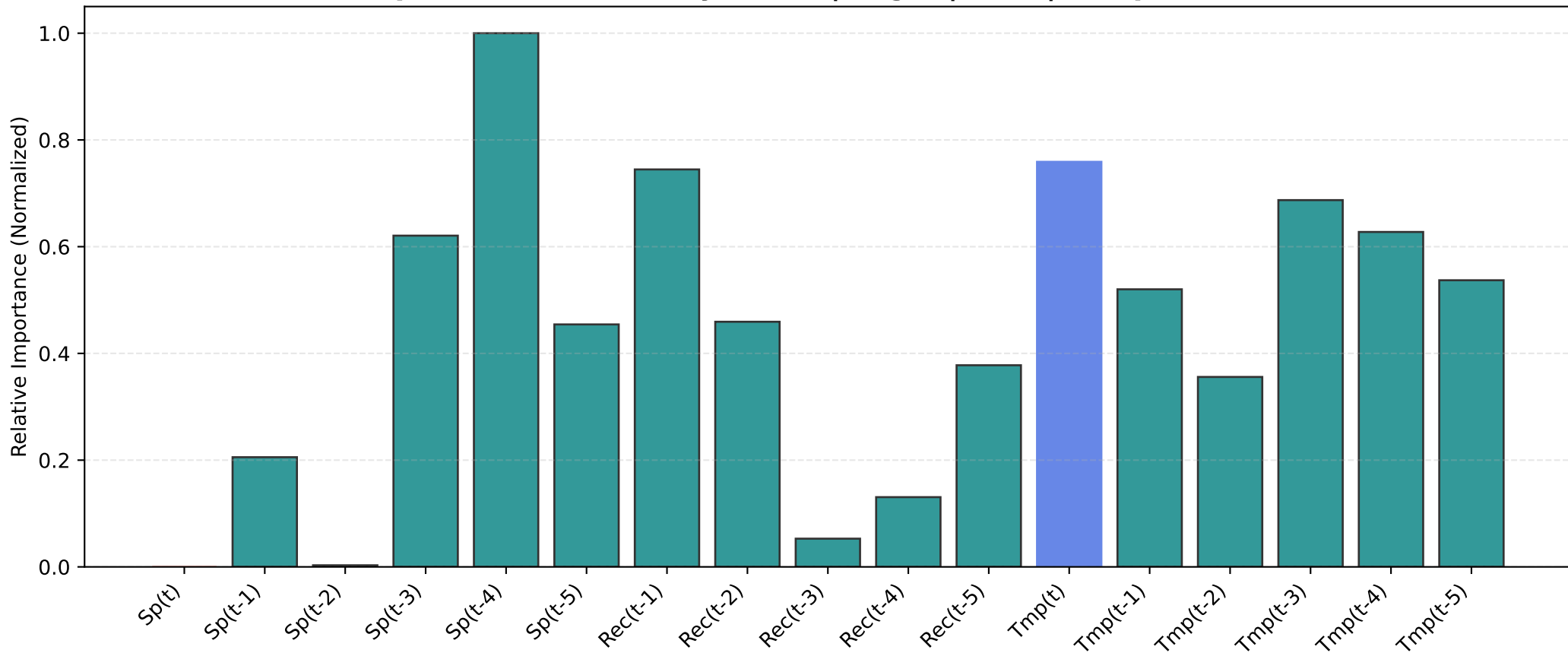
## Training & Validation Loss



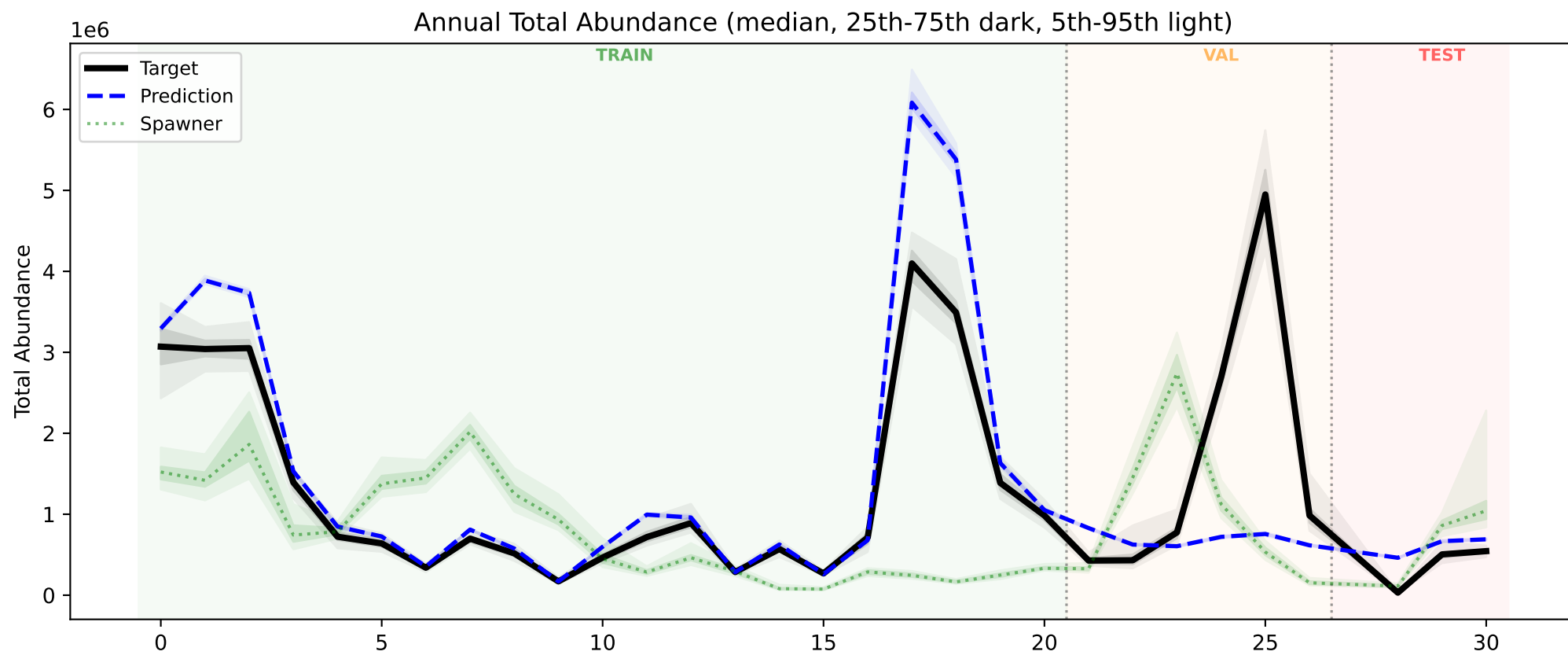
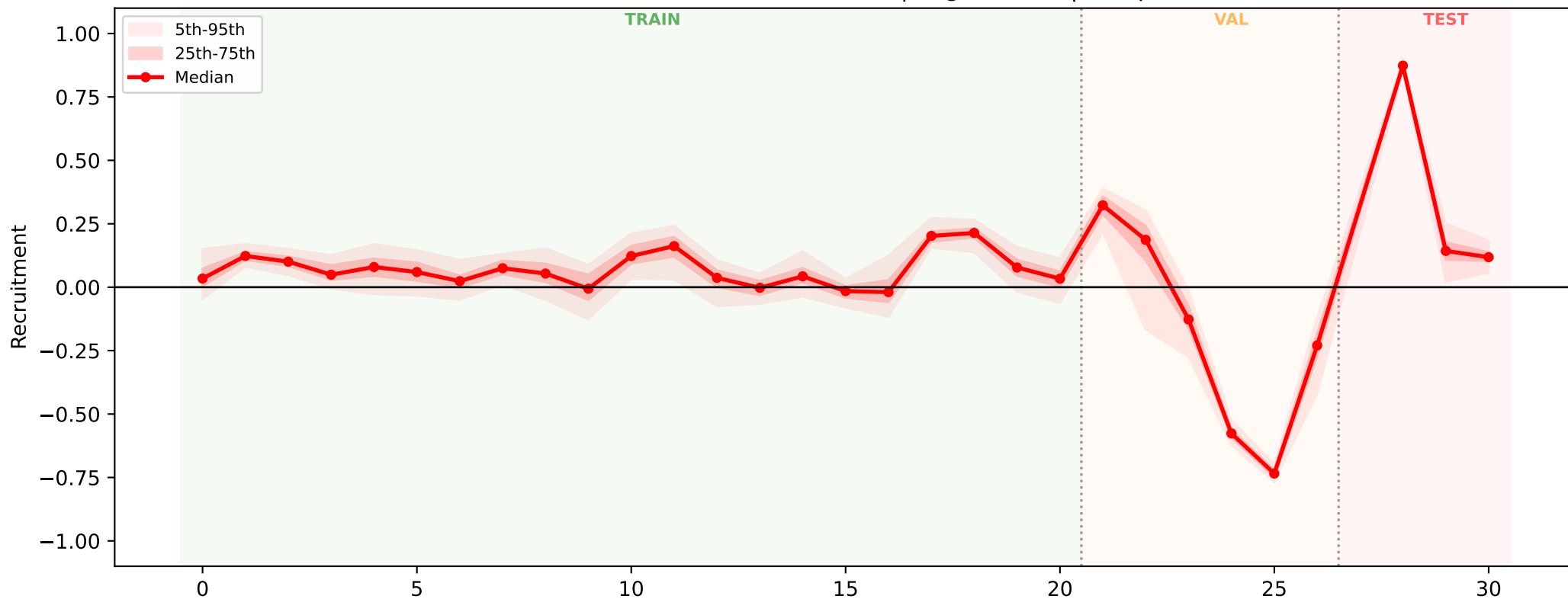
## OneCycleLR Schedule



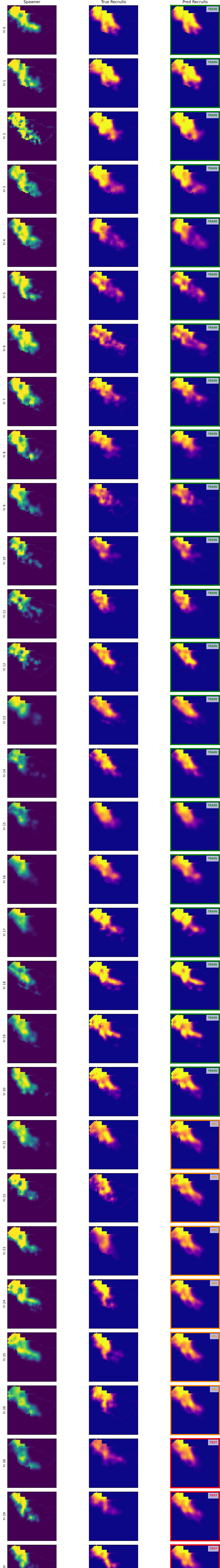
**Input Channel Priority: REAL | Lag 5 | MSE | w/ Spawner(t)**



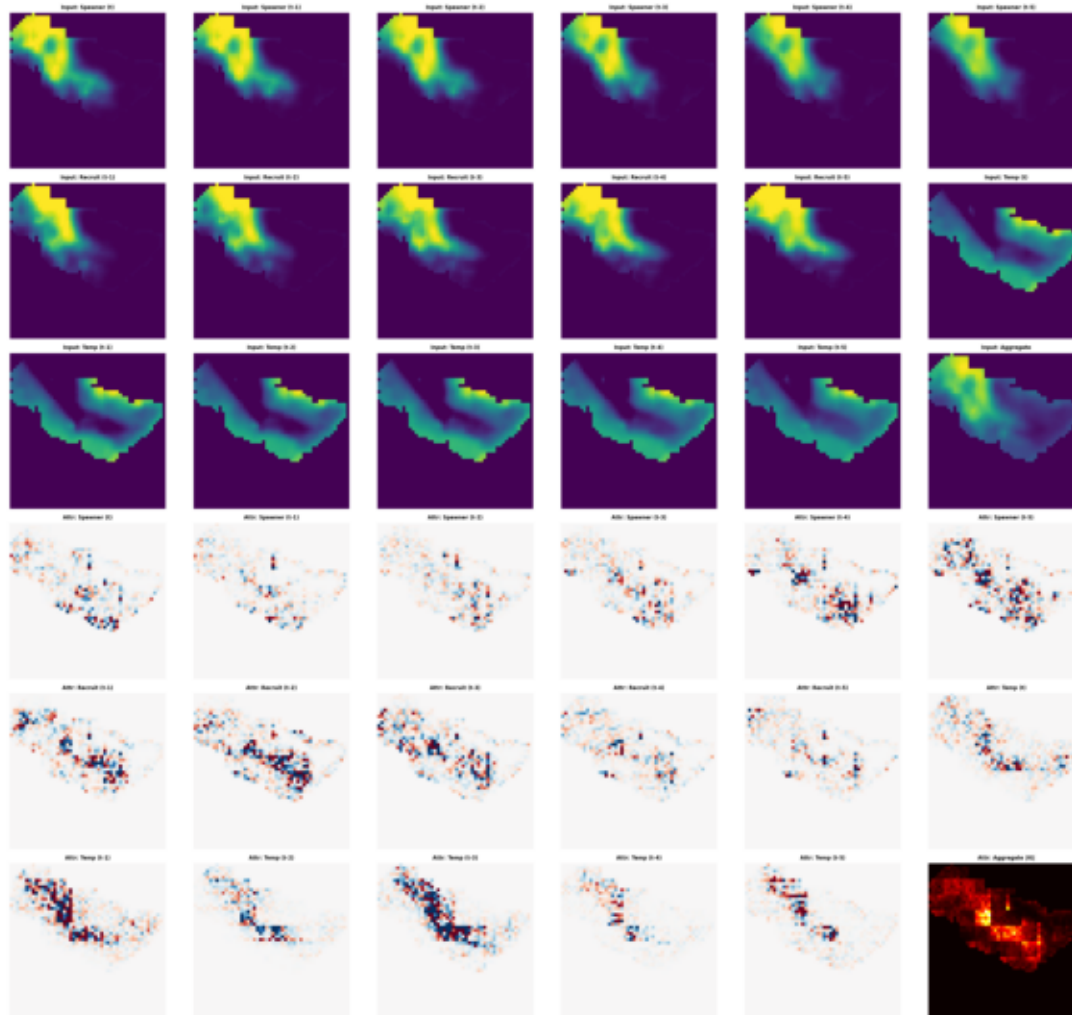
Annual Recruitment Deviations: REAL | Lag 5 (MSE) | w/ Spawner(t)



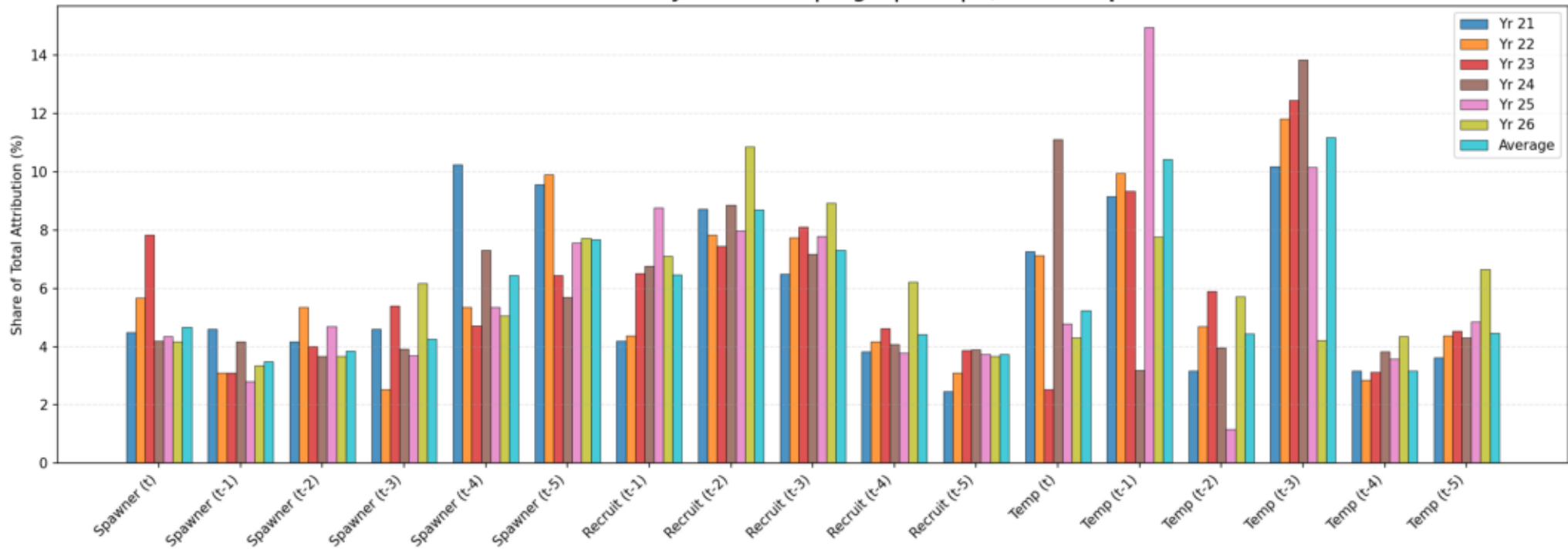
Spatial Trajectory: REAL | Lag 5 | MSE | w/ Spawner(t)



Year AVG -- REAL | Lag 5 | MSE | w/ Current Spawner  
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)

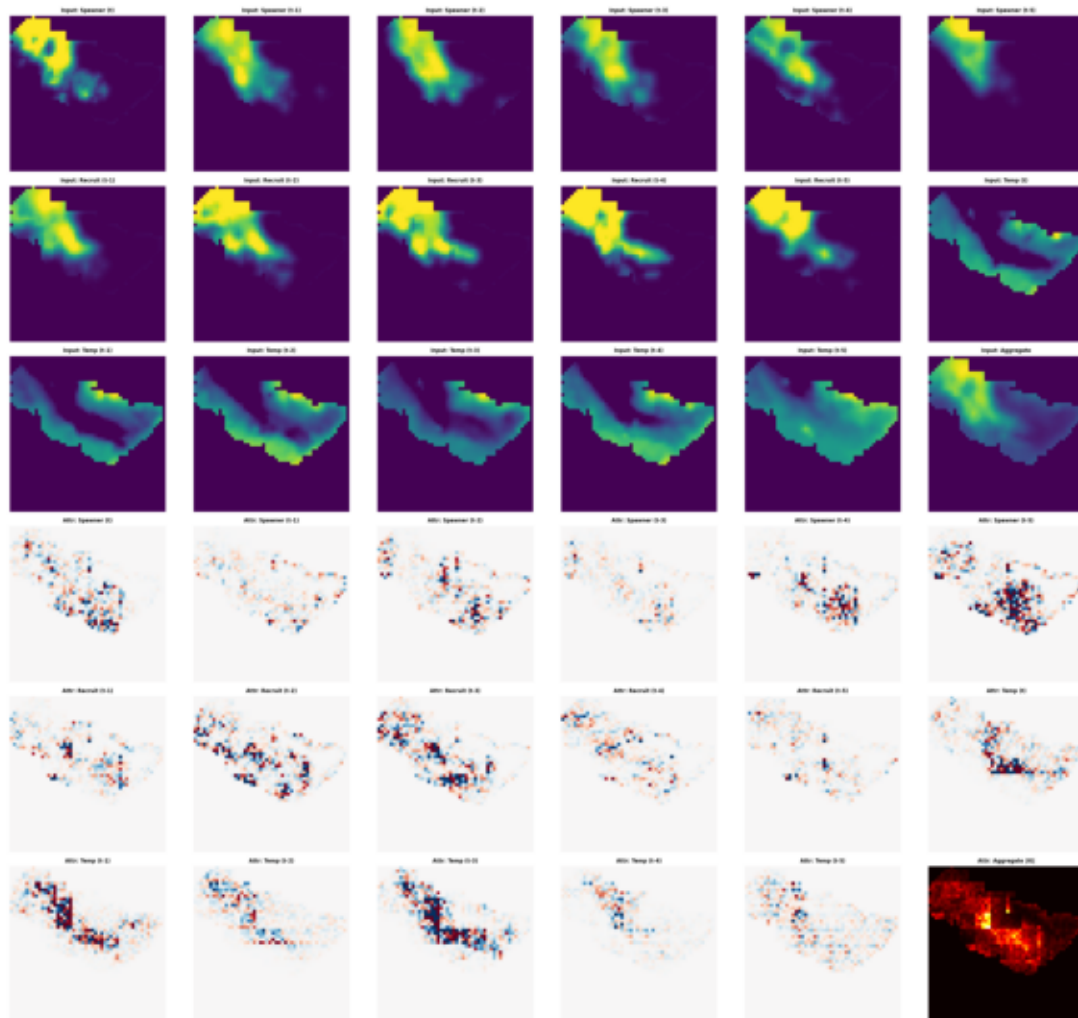


Channel Attribution by Year: REAL | Lag 5 | MSE | w/ Current Spawner

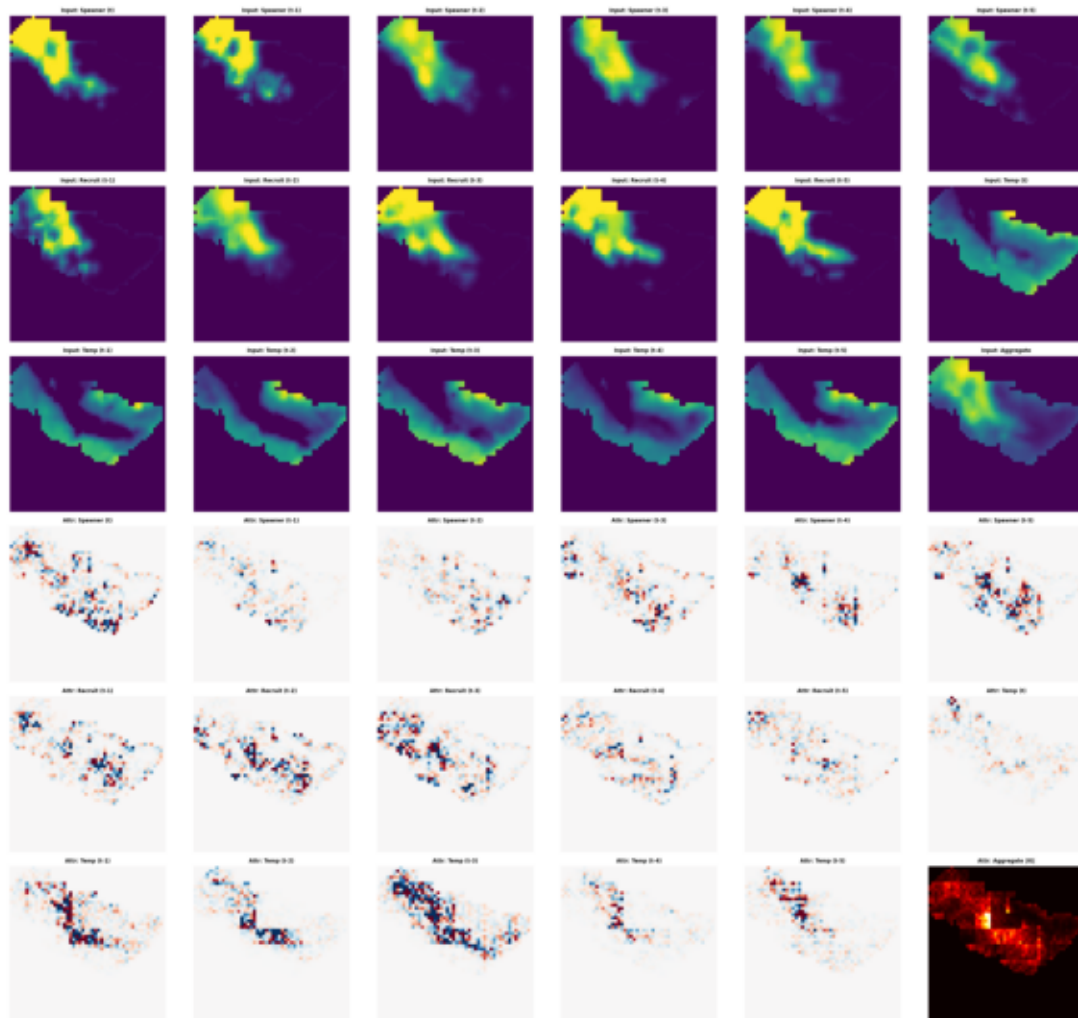




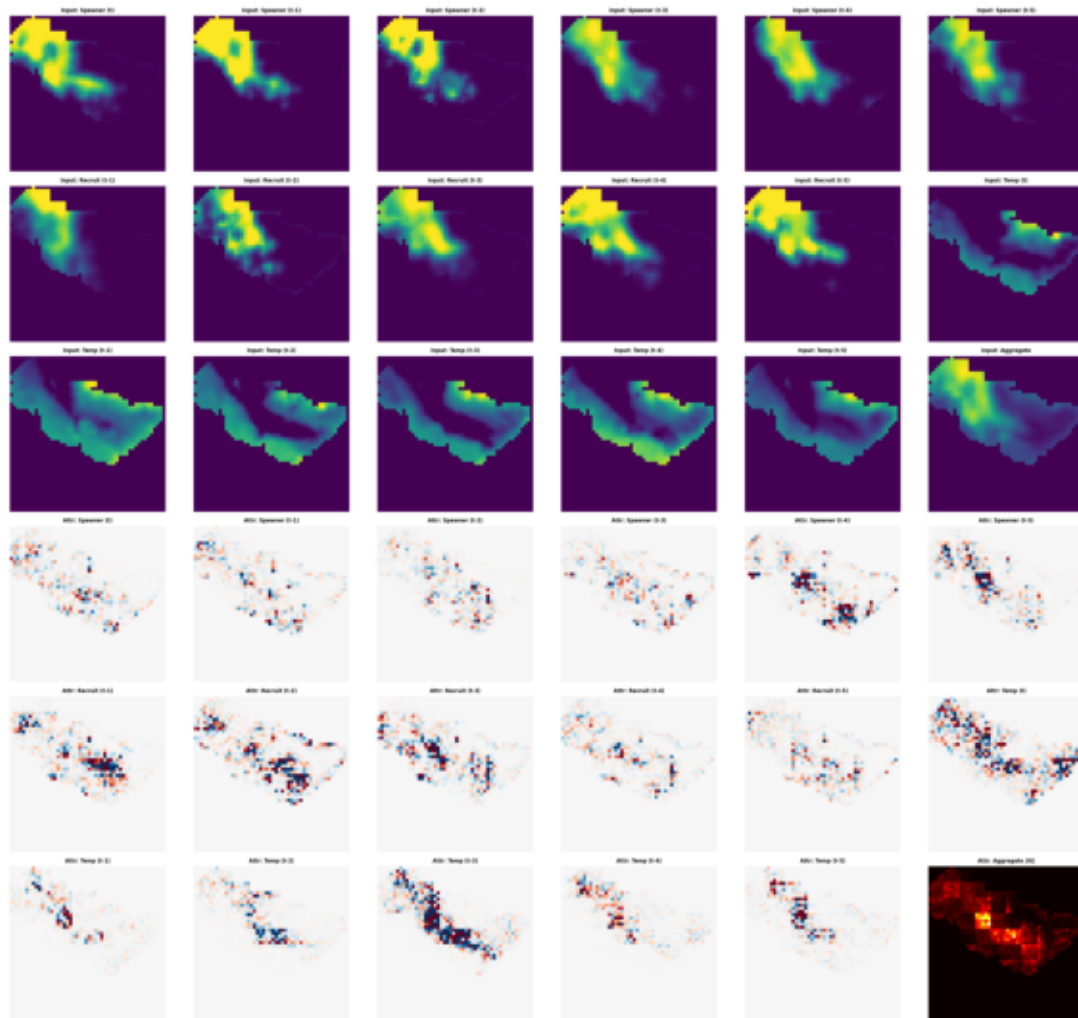
Year 22 — REAL | Lag 5 | MSE | w/ Current Spawner  
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



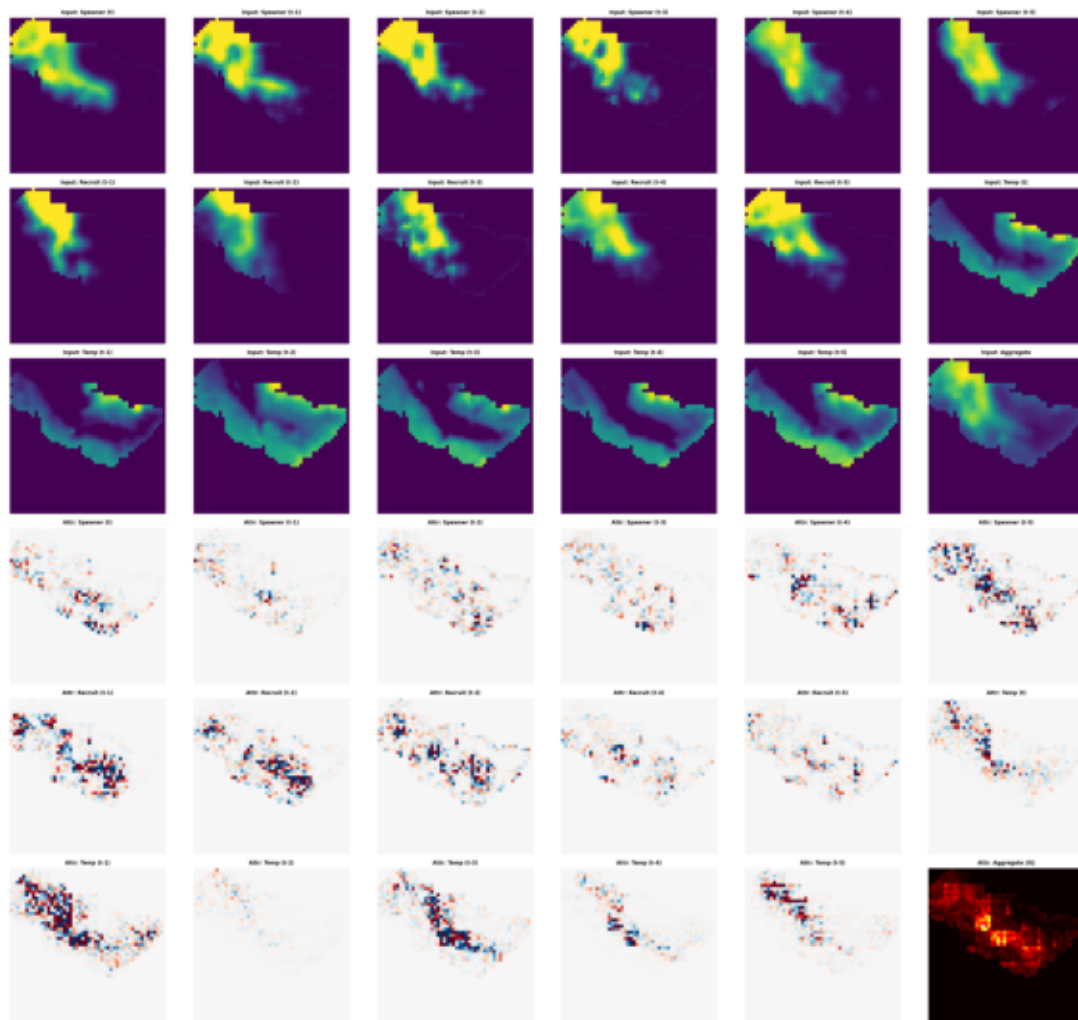
Year 23 — REAL | Lag 5 | MSE | w/ Current Spawner  
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



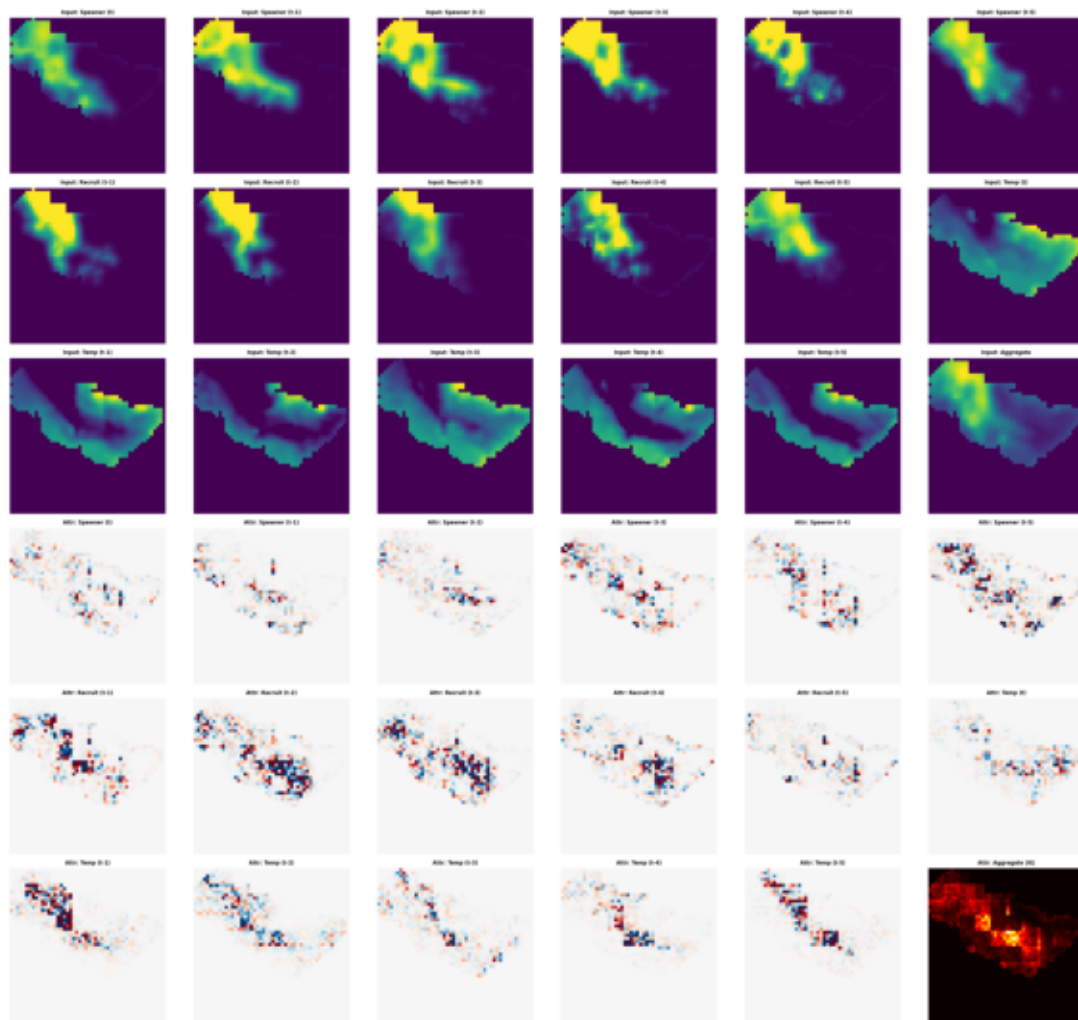
Year 24 — REAL | Lag 5 | MSE | w/ Current Spawner  
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 25 — REAL | Lag 5 | MSE | w/ Current Spawner  
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 26 — REAL | Lag 5 | MSE | w/ Current Spawner  
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Summary Statistics: REAL | Lag 5 | MSE | w/ Spawner(t)

| Phase | Pred_Mean | Tgt_Mean | Pred_SD | Pred_P50 | Pred_P95 | Abundance_Capture | MAE    | Spearman |
|-------|-----------|----------|---------|----------|----------|-------------------|--------|----------|
| TRAIN | 1854.4    | 1480.3   | 8012.0  | 0.0      | 9885.0   | 1.253             | 543.7  | 0.978    |
| VAL   | 780.0     | 1950.8   | 2171.9  | 0.0      | 5426.2   | 0.400             | 1711.6 | 0.826    |
| TEST  | 683.4     | 406.3    | 2293.8  | 0.0      | 4226.0   | 1.682             | 453.5  | 0.825    |

Detailed Metrics (R-style): REAL | Lag 5 | MSE | w/ Spawner(t)

| Phase | RMSE   | RMSE_unbias | Bias    | Bias_% | Spearman | Pearson | Pct_Error | Zero_F1 | TopK_Acc | TopK_Jaccard | Quant_Err | Skill_MSE | Skill_MAE |
|-------|--------|-------------|---------|--------|----------|---------|-----------|---------|----------|--------------|-----------|-----------|-----------|
| TRAIN | 1641.5 | 1593.7      | 374.1   | 3.3%   | 0.987    | 0.986   | 15.2%     | 0.989   | 0.943    | 0.893        | 0.12      | 0.891     | 0.853     |
| VAL   | 5977.0 | 5806.6      | -1170.8 | 4.0%   | 0.832    | 0.721   | -16.6%    | 0.895   | 0.670    | 0.513        | 0.90      | 0.063     | 0.417     |
| TEST  | 1658.1 | 1629.6      | 277.1   | 3.4%   | 0.833    | 0.674   | 497.2%    | 0.912   | 0.720    | 0.596        | 0.81      | -133.848  | -2.400    |